

Bureau of Indian Standards (BIS) Technical Knowledge Corpus

Exhaustive Clause-by-Clause Reference Manual & High-Density RAG Knowledge Base for SIH 2026

1. IS 456 : 2000 — Plain and Reinforced Concrete - Code of Practice (Fourth Revision)

Standard Metadata: Committee: CED 2 (Cement & Concrete) | Scope: Design, construction, materials, durability, quality control, and testing of structural concrete across India.

1.1 Scope, Materials & Mandatory Specifications

IS 456 deals with the general structural use of plain and reinforced concrete in buildings and civil engineering structures. It is not directly applicable to special structures like bridges, chimneys, dams, or pre-stressed concrete, which are governed by dedicated sister codes (such as IRC 112 or IS 1343).

- Cement Compliance:** Mandates compliance with standard cement specifications: 33-Grade OPC (IS 269), 43-Grade OPC (IS 8112), 53-Grade OPC (IS 12269), Portland Slag Cement (IS 455), Portland Pozzolana Cement (IS 1489 Parts 1 and 2), and Sulphate Resisting Portland Cement (IS 12330).
- Aggregates:** Aggregates must strictly adhere to IS 383:2016. Nominal maximum size of coarse aggregate is typically 20 mm for structural members, though 10 mm or 40 mm may be permitted depending on spacing and clear cover constraints.
- Reinforcement:** Reinforcing bars must comply with IS 432 (Part 1) for Mild Steel and Medium Tensile Steel bars (Fe 250), or IS 1786 for High Strength Deformed Steel Bars and wires (Fe 415, Fe 500, Fe 550, and Fe 600).
- Water Quality Criteria (Clause 5.4):** Mixing and curing water must be clean and free from harmful amounts of oils, acids, alkalis, salts, and organic materials. Permissible limits of solids per IS 456 Clause 5.4 Table 1:
 - Organic solids: Maximum 200 mg/L
 - Inorganic solids: Maximum 3000 mg/L
 - Sulphates (as SO3): Maximum 400 mg/L
 - Chlorides (as Cl): Maximum 2000 mg/L for plain concrete; maximum 500 mg/L for reinforced concrete work
 - Suspended matter: Maximum 2000 mg/L
 - pH Value: Must not be less than 6.0.

1.2 Durability Requirements & Environmental Exposure Classes (Clause 8 & Table 2)

IS 456 links structural durability directly to environmental exposure severity. Concrete proportioning must satisfy the dual criteria of minimum compressive grade, minimum cementitious content, and maximum permissible water-cement ratio.

Exposure Environment	Exposure Description / Environment	Min RCC Grade	Min Cement (kg/m³)	Max Free W/C Ratio	Min Clear Cover (mm)
Mild	Concrete surfaces protected against weather or aggressive conditions (interior rooms).	M20	300	0.55	20 mm (15 mm if bars ≤ 12mm)
Moderate	Concrete sheltered from severe rain/freezing; concrete continuously underwater; concrete in non-aggressive soil.	M25	300	0.50	30 mm
Severe	Surfaces exposed to severe rain, alternate wetting/drying, coastal atmosphere, or buried in corrosive soils.	M30	320	0.45	45 mm
Very Severe	Concrete exposed to seawater spray, corrosive fumes, or severe freezing conditions while wet.	M35	340	0.45	50 mm
Extreme	Members in tidal zones, splash zones of marine structures, or direct contact with aggressive chemicals.	M40	360	0.40	75 mm

1.3 Concrete Mechanical Properties & Mathematical Formulations

- Characteristic Compressive Strength (f_{ck}):** Compressive strength of 150 mm cubes at 28 days cured in water below which not more than 5 percent of the test results are expected to fall.
- Flexural Tensile Strength (Modulus of Rupture - Clause 6.2.2):**

$$f_{cr} = 0.7 * \sqrt{f_{ck}} \text{ [N/mm}^2 \text{ or MPa]}$$

- Static Modulus of Elasticity (E_c - Clause 6.2.3.1):**

$$E_c = 5000 * \sqrt{f_{ck}} \text{ [N/mm}^2 \text{ or MPa]}$$

Actual measured values may differ by ±20 percent depending on aggregate properties.

- **Shrinkage & Creep:** Total shrinkage strain of concrete for design purposes in building structures is taken as 0.0003 . Creep coefficient values ($\epsilon_{\text{creep}} = \epsilon_{\text{creep strain}} / \epsilon_{\text{elastic strain}}$): 7 days = 2.2 ; 28 days = 1.6 ; 1 year = 1.1 .

1.4 Limit State Method (LSM) Design Principles (Clause 35–40)

IS 456 structures must satisfy Limit State of Collapse (flexure, compression, shear, torsion) and Limit State of Serviceability (deflection, cracking).

- **Partial Safety Factors for Materials (γ_m - Clause 36.4.2):**
 - Concrete in collapse: $\gamma_{mc} = 1.50$
 - Steel in collapse: $\gamma_{ms} = 1.15$
 - Serviceability limit states: $\gamma_m = 1.00$ for both concrete and steel.
- **Design Compressive Stress-Strain Curve for Concrete:** Parabolic up to strain of 0.002 , then constant until ultimate failure strain of 0.0035 . Maximum design stress is taken as:

$$\text{Design Stress} = (0.67 * f_{ck}) / 1.50 = 0.446 * f_{ck}$$

- **Limiting Depth of Neutral Axis ($x_{u,max}/d$):**

- For Mild Steel Fe 250: $x_{u,max}/d = 0.53$
- For HYSD Fe 415: $x_{u,max}/d = 0.48$
- For HYSD Fe 500: $x_{u,max}/d = 0.46$
- For HYSD Fe 550: $x_{u,max}/d = 0.44$

- **Limiting Moment of Resistance ($M_{u,lim}$):**

$$M_{u,lim} = 0.36 * f_{ck} * b * x_{u,max} * (d - 0.42 * x_{u,max})$$

For Fe 415: $M_{u,lim} = 0.138 \cdot f_{ck} \cdot b \cdot d^2$; For Fe 500: $M_{u,lim} = 0.133 \cdot f_{ck} \cdot b \cdot d^2$.

- **Shear Design (Clause 40):** Nominal shear stress $\tau_v = V_u / (b \cdot d)$. Design shear strength τ_{vc} depends on concrete grade and percentage of tensile steel $\rho_{st} = 100 A_{st} / (b \cdot d)$. Maximum shear stress is capped at $\tau_{c,max}$ per Table 20 (e.g., 2.8 MPa for M20, 3.1 MPa for M25, 3.5 MPa for M30, 4.0 MPa for M40).
- **Torsion Design (Clause 41):** Equivalent shear $V_e = V_u + 1.6 (T_u / b)$; Equivalent bending moment $M_{e1} = M_u + M_t$, where $M_t = T_u (1 + D/b) / 1.7$.

1.5 Structural Detailing, Spacing & Cover Specifications (Clause 26)

- **Minimum Longitudinal Reinforcement in Beams:** $A_{s,min} / (b \cdot d) = 0.85 / f_y$.
- **Maximum Longitudinal Reinforcement in Beams:** $A_{s,max} = 0.04 \cdot b \cdot D$ (4% of gross cross-sectional area).
- **Columns:** Minimum longitudinal steel is 0.8% of gross area; maximum is 6% (or 4% where splices are required). Minimum diameter of longitudinal bars is 12 mm . Minimum of 4 bars in rectangular columns and 6 bars in circular columns.
- **Spacing of Transverse Ties in Columns:** Pitch must not exceed: (a) Least lateral dimension of the column, (b) 16 times smallest diameter of longitudinal bar, or (c) 300 mm .
- **Nominal Clear Covers (Table 16):** Footings: 50 mm ; Columns: 40 mm ; Beams: 25 mm ; Slabs: 20 mm .

RAG Keyword Vector Seed: IS 456, M20, M25, M30, exposure classes, water-cement ratio, clear cover, $0.7 \sqrt{f_{ck}}$, $5000 \sqrt{f_{ck}}$, partial safety factor, 1.5 concrete, 1.15 steel, limiting neutral axis, shear design, τ_{vc} , stirrups, column reinforcement, $0.85/f_y$.

2. IS 800 : 2007 — General Construction in Steel - Code of Practice (Third Revision)

Standard Metadata: Committee: CED 7 (Structural Engineering & Structural Sections) | Scope: Limit State Method design of hot-rolled structural steel sections, cold-formed sections, composite elements, bolted & welded connections.

2.1 Scope, Material Properties & Design Philosophies

IS 800:2007 adopted the Limit State Design (LSD) methodology, aligning Indian structural steel engineering with international Eurocode and AISC codes. The Working Stress Method (WSM) is preserved only as an alternative design procedure in Annex B.

- Structural Steel Standard:** Structural steel must conform to IS 2062:2011 (Grades E250, E300, E350, E410, E450). Properties for standard structural steel (E250):
 - Modulus of Elasticity (\$E\$): $2.0 \times 10^5 \text{ N/mm}^2$ (200 GPa)
 - Shear Modulus (\$G\$): $0.77 \times 10^5 \text{ N/mm}^2$ (77 GPa)
 - Poisson's Ratio (\$\mu\$): 0.30 in elastic range; 0.50 in plastic range
 - Coefficient of Thermal Expansion (\$\alpha\$): $12 \times 10^{-6} / ^\circ\text{C}$
 - Unit Mass (\$\rho\$): 7850 kg/m^3 (78.5 kN/m^3).
- Classification of Cross-Sections (Clause 3.7.2 & Table 2):**
 - Class 1 (Plastic):* Can form plastic hinges and achieve full plastic moment capacity with required rotation capacity.
 - Class 2 (Compact):* Can develop full plastic moment resistance \$M_p\$, but local buckling prevents full rotational plastic hinge capacity.
 - Class 3 (Semi-Compact):* Stress in extreme compression fibers reaches yield stress \$f_y\$, but local buckling prevents development of plastic moment.
 - Class 4 (Slender):* Local buckling occurs in compression elements before yield stress is achieved.

2.2 Maximum Slenderness Ratios (\$\lambda = KL/r\$) — Clause 3.8 & Table 3

Effective slenderness ratio \$\lambda = KL/r\$ dictates the member stability against flexural and torsional buckling.

Member Category and Load Type	Max \$\lambda = KL/r\$
Member carrying compressive loads resulting from dead loads and imposed loads (Columns/Struts)	180
Member subjected to compression loads resulting only from wind or earthquake forces	250
Member subjected to compressive loads resulting from wind/seismic forces during erection	350
Member normally acting as a tie in a roof truss, but subjected to reversal of stress due to wind or earthquake	350
Member always in pure tension (excluding pretensioned ties and cables)	400
Lacing bars in built-up compression members (Clause 7.6.3)	145

2.3 Tension Member Design (Clause 6)

The design strength of a tension member \$T_d\$ is taken as the lowest of:

- Gross Section Yielding (\$T_{dg}\$ - Clause 6.2):**

$$T_{dg} = (A_g * f_y) / \gamma_{m0} = (A_g * f_y) / 1.10$$

- Net Section Rupture (\$T_{dn}\$ - Clause 6.3):**

- For plates with holes: $T_{dn} = (0.9 \cdot A_n \cdot f_u) / \gamma_{m1} = (0.9 \cdot A_n \cdot f_u) / 1.25$.
- For single angle connected through one leg:

$$T_{dn} = (0.9 * A_{nc} * f_u) / \gamma_{m1} + (\eta * A_{go} * f_y) / \gamma_{m0}$$

where shear lag factor \$ \eta = 1.4 - 0.076 (w/t)(f_y/f_u)(b_s/L_c) \le (f_u / \gamma_{m0}) / (f_y / \gamma_{m1}) \ge 0.7\$.

- Block Shear Failure (\$T_{db}\$ - Clause 6.4):** Evaluates tear-out along shear planes and tensile rupture along normal planes.

2.4 Compression Member Design (Clause 7)

- Design Compressive Strength (\$P_d\$):**

$$P_d = A_e \cdot f_{cd}$$

where f_{cd} is the design compressive stress:

$$f_{cd} = (f_y / \gamma_{m0}) / [\phi + (\phi^2 - \lambda_e^2)^{0.5}]$$

where $\phi = 0.5 [1 + \alpha (\lambda_e - 0.2) + \lambda_e^2]$ and non-dimensional slenderness $\lambda_e = \sqrt{f_y / f_{cc}}$.

• **Buckling Classes (a, b, c, d - Table 7):** Imperfection factor α :

- Curve a: $\alpha = 0.21$ (Rolled I-sections strong axis)
- Curve b: $\alpha = 0.34$ (Rolled I-sections weak axis, RHS, SHS)
- Curve c: $\alpha = 0.49$ (Welded I-sections, Channels, Angles)
- Curve d: $\alpha = 0.76$ (Thick welded flanges > 40 mm, built-up members).

2.5 Connections: Fasteners & Welds (Clause 10)

• **Partial Safety Factors for Connections (γ_m):**

- Bearing bolts (γ_{mb}): 1.25
- Friction grip bolts (slip resistance γ_{mf}): 1.10 for normal loads; 1.25 for collapse
- Welds in workshop / fabrication shop (γ_{mw}): 1.25
- Welds made on construction site / field (γ_{mw}): 1.50.

• **Shear Capacity of Bearing Bolt (V_{dsb} - Clause 10.3.3):**

$$V_{dsb} = [(f_{ub} / \sqrt{3}) \cdot (n_n \cdot A_{nb} + n_s \cdot A_{sb})] / 1.25$$

• **Fillet Weld Design Strength (f_{wd} - Clause 10.5.7):**

$$f_{wd} = f_{wn} / \gamma_{mw} = [f_u / (\sqrt{3} \cdot \gamma_{mw})]$$

Effective throat thickness $t = k \cdot s$, where $k = 0.70$ for 60° – 90° fusion angle. Minimum size of fillet weld is 3 mm for plates up to 10 mm thickness, 5 mm for 10–20 mm, and 6 mm for 20–32 mm.

RAG Keyword Vector Seed: IS 800, slenderness ratio, 180 compression, 350 tension reversal, 400 pure tension, γ_{m0} 1.10, γ_{m1} 1.25, γ_{mw} 1.25 shop 1.50 field, buckling curves a b c d, shear lag beta, fillet weld throat 0.7s, bolt shear V_{dsb} .

3. IS 875 (Parts 1–5) : 1987 — Code of Practice for Design Loads (Other than Earthquake)

Standard Metadata: Committee: CED 37 (Structural Safety & Design Provisions) | Scope: Dead, Imposed (Live), Wind, Snow, and Combination Design Loads for all buildings and civil engineering works in India.

3.1 Part 1 — Dead Loads (Unit Weights of Building Materials)

IS 875 Part 1 specifies the standard unit weights (kN/m^3) for building structural and finishing components to be used in dead load calculations:

- Plain Cement Concrete (PCC): 24.0 kN/m^3
- Reinforced Cement Concrete (RCC): 25.0 kN/m^3 (assumes 1% reinforcement steel)
- Structural Steel: 78.5 kN/m^3 (7850 kg/m^3)
- Common Burnt Clay Brick Masonry (cement mortar): $18.8 \text{ to } 19.5 \text{ kN/m}^3$
- Autoclaved Aerated Concrete (AAC) Blocks: $5.5 \text{ to } 7.5 \text{ kN/m}^3$
- Plaster (Cement plaster): $20.0 \text{ to } 21.0 \text{ kN/m}^3$
- Floor Finishes (Tiles / Screed): 22.0 kN/m^3
- Timber (Teak / Sal): $6.5 \text{ to } 8.5 \text{ kN/m}^3$.

3.2 Part 2 — Imposed Loads (Live Loads for Occupancies)

Occupancy Classification	Uniformly Distributed Load (UDL)	Concentrated Point Load
Residential: Living rooms, bedrooms, kitchen	2.0 kN/m ²	1.8 kN
Residential: Bathrooms, toilets	2.0 kN/m ²	—
Residential: Corridors, common staircases, balconies	3.0 kN/m ²	1.5 kN
Offices: General work rooms with light filing equipment	2.5 kN/m ²	2.7 kN
Offices: Banking halls, conference rooms, public areas	3.0 kN/m ²	2.7 kN
Commercial / Retail: General shopping areas	4.0 kN/m ²	3.6 kN
Educational: Classrooms and lecture halls	3.0 kN/m ²	2.7 kN
Warehouses / Heavy Industrial Storage	5.0 to 10.0 kN/m ²	4.5 to 7.0 kN
Roofs: Inaccessible flat roof (access for maintenance only)	0.75 kN/m ²	1.5 kN
Roofs: Accessible terrace / flat roof	1.5 kN/m ²	3.75 kN

3.3 Part 3 — Wind Loads (As per Revised IS 875 Part 3 : 2015)

Wind forces act as fluctuating aerodynamic pressures and are evaluated mathematically through basic wind speed maps, terrain exposure factors, and gust effects.

- **Basic Wind Speed (V_b):** Peak gust wind speed averaged over 3 seconds at 10 m height above ground in open terrain, with a 50-year return period. Ranges from 33 m/s (central plateau like Bengaluru), 39 m/s (Maharashtra/Jalgaon), 44 m/s (Delhi/NCR), 47 m/s (Mumbai/Kolkata), up to 50–55 m/s (cyclone-prone eastern seaboard of Odisha and Andhra Pradesh).
- **Design Wind Speed (V_z at height z):**

$$V_z = V_b * k_1 * k_2 * k_3 * k_4 \text{ [m/s]}$$

- k_1 : Probability factor (risk coefficient). $k_1 = 1.0$ for standard 50-year life structures; 1.07 for important buildings (hospitals/communication towers).
- k_2 : Terrain roughness, height, and structure size factor. Categories 1 (open sea/desert) to 4 (dense city centers with closely spaced tall buildings). Increases with elevation.
- k_3 : Topography factor. Accounts for wind acceleration over ridges, hills, and escarpments ($k_3 = 1.0$ for slopes $< 3^\circ$, up to 1.36 on crests).
- k_4 : Importance factor for cyclonic regions ($k_4 = 1.30$ for post-cyclone emergency facilities; 1.15 for industrial structures on coast; 1.00 for inland).
- **Design Wind Pressure (p_z):**

$$p_z = 0.6 * (V_z)^2 \text{ [N/m}^2\text{]}$$

• **Wind Force on Structure (\$F\$):**

$$F = (C_{pe} - C_{pi}) * A * p_z$$

where C_{pe} is external pressure coefficient, C_{pi} is internal pressure coefficient (± 0.2 for normal permeability; ± 0.5 for medium openings), and A is effective surface area.

3.4 Part 4 — Snow Loads

Governs structures situated in northern/alpine zones. Snow load on roof $S = \mu \cdot S_0$, where S_0 is ground snow load in kN/m^2 , and μ is the shape coefficient accounting for roof pitch, curvature, and thermal insulation loss.

3.5 Part 5 — Load Combinations & Limit State Factors

• **Limit State of Collapse Combinations:**

- $1.5 (DL + IL)$
- $1.2 (DL + IL \pm WL \text{ ext{ or } EL})$
- $1.5 (DL \pm WL \text{ ext{ or } EL})$
- $0.9 DL \pm 1.5 (WL \text{ ext{ or } EL})$ *(mandatory stability load check against overturning, slide, and tension uplift)*.

• **Limit State of Serviceability Combinations:**

- $1.0 DL + 1.0 IL$
- $1.0 DL + 0.8 IL \pm 0.8 (WL \text{ ext{ or } EL})$
- $1.0 DL \pm 1.0 (WL \text{ ext{ or } EL})$.

RAG Keyword Vector Seed: IS 875, dead loads, RCC 25 kN/m³, residential live load 2.0 kN/m², wind speed V_b , $V_z = V_b * k_1 * k_2 * k_3 * k_4$, $p_z = 0.6 * V_z^2$, terrain category k_2 , cyclonic factor k_4 , collapse load combinations $1.5(DL+IL)$, $1.2(DL+IL+WL)$, $0.9DL+1.5WL$ uplift.

4. IS 1893 (Part 1) : 2016 — Criteria for Earthquake Resistant Design of Structures

Standard Metadata: Committee: CED 39 (Earthquake Engineering) | Scope: Seismic hazard evaluation, seismic zoning map of India, dynamic base shear calculation, lateral drift limits, and torsional irregularity checks.

4.1 Seismic Philosophy & Seismic Zoning of India (Clause 6)

The design earthquake forces specified in IS 1893 are significantly smaller than the actual forces that would develop if the structure were to remain purely linear-elastic during a Maximum Considered Earthquake (MCE). The code relies on **structural ductility, energy dissipation, and plastic hinge formation** to prevent progressive collapse.

• Seismic Zone Categorization (Zone Factor Z — Table 3):

- **Zone II (Low Seismic Intensity):** $Z = 0.10$ (Southern Peninsula, parts of central India).
- **Zone III (Moderate Seismic Intensity):** $Z = 0.16$ (Western India, Mumbai, Jalgaon, Chennai, parts of UP/Punjab).
- **Zone IV (Severe Seismic Intensity):** $Z = 0.24$ (Delhi/NCR, Northern plains, parts of Gujarat, foothills of Himalayas).
- **Zone V (Very Severe Seismic Intensity):** $Z = 0.36$ (Entire North-Eastern states, Kashmir, Himachal Pradesh, Uttarakhand, Rann of Kutch, Andaman & Nicobar).
- *Zone I was completely abolished in the 2002 revision.*

4.2 Seismic Weight Calculation (W — Clause 7.4)

The total seismic weight of the building W is the sum of seismic weights across all floors:

- 100% of all Dead Loads (DL) including self-weight, masonry walls, finishes, and permanent MEP equipment.
- Percentage of Imposed Live Load (IL) per Table 10:
 - For live loads $\leq 3.0 \text{ kN/m}^2$: Include 25% of live load.
 - For live loads $> 3.0 \text{ kN/m}^2$: Include 50% of live load.
 - Roof live loads: 0% (no live load is counted on roofs during an earthquake).

4.3 Design Horizontal Base Shear Formulation (V_B) — Clause 7.2

The total design lateral force at the base of the structure:

$$V_B = A_h \cdot W$$

where A_h is the horizontal seismic acceleration coefficient:

$$A_h = (Z / 2) \cdot (I / R) \cdot (S_a / g)$$

- **$Z/2$:** Divides the Maximum Considered Earthquake (MCE) by 2 to yield the Design Basis Earthquake (DBE).
- **Importance Factor (I — Table 8):**
 - Important service/lifeline buildings (Hospitals, schools, disaster management rooms, telecommunications): $I = 1.5$
 - Residential/commercial with occupancy > 200 persons: $I = 1.2$
 - All other regular residential buildings: $I = 1.0$.
- **Response Reduction Factor (R — Table 9):** Represents ductility and energy absorption capacity:
 - Special Moment Resisting Frame (SMRF) with ductile detailing per IS 13920: $R = 5.0$
 - Ordinary Moment Resisting Frame (OMRF): $R = 3.0$ (prohibited in Zones IV & V)
 - Steel frames with concentric braces: $R = 4.0$; with eccentric braces: $R = 5.0$.
- **Spectral Acceleration Coefficient (S_a/g — Clause 6.4.2):** Function of fundamental period T_a and soil type (5% damping):
 - *Type I (Rock or Hard Soil):* $S_a/g = 1 + 15 T_a$ ($0 \leq T_a \leq 0.10 \text{ s}$); $S_a/g = 2.50$ ($0.10 \leq T_a \leq 0.40 \text{ s}$); $S_a/g = 1.00 / T_a$ ($T_a > 0.40 \text{ s}$).
 - *Type II (Medium Soil):* $S_a/g = 1 + 15 T_a$ ($0 \leq T_a \leq 0.10 \text{ s}$); $S_a/g = 2.50$ ($0.10 \leq T_a \leq 0.55 \text{ s}$); $S_a/g = 1.36 / T_a$ ($T_a > 0.55 \text{ s}$).
 - *Type III (Soft Soil):* $S_a/g = 1 + 15 T_a$ ($0 \leq T_a \leq 0.10 \text{ s}$); $S_a/g = 2.50$ ($0.10 \leq T_a \leq 0.67 \text{ s}$); $S_a/g = 1.67 / T_a$ ($T_a > 0.67 \text{ s}$).

4.4 Natural Period & Storey Drift Limitations

• Fundamental Natural Period (T_a — Clause 7.6.2):

- Bare RCC frame without infill: $T_a = 0.075 \cdot h^{0.75}$

- Bare Steel frame without infill: $T_a = 0.085 \cdot h^{0.75}$
- Buildings with full masonry infill panels: $T_a = (0.09 \cdot h) / \sqrt{d}$ (where h is height in meters, d is base dimension along earthquake direction in meters).
- **Lateral Storey Drift Limitation (Clause 7.11.1):** The storey drift under design lateral force with partial safety factor of 1.0 must not exceed **0.004 times the storey height** ($\Delta / h_i \leq 0.004$). For a 3.0 m (3000 mm) storey, max lateral drift is 12 mm.

RAG Keyword Vector Seed: IS 1893:2016, Seismic Zones II III IV V, Zone factors 0.10 0.16 0.24 0.36, Base shear $V_B = A_h \cdot W$, $A_h = (Z/2) \cdot (I/R) \cdot (S_a/g)$, SMRF $R=5$, OMRF $R=3$, Importance factor $I=1.5$, seismic weight 25% 50% LL, drift limit 0.004h, T_a period.

5. IS 3043 : 2018 — Code of Practice for Earthing (First Revision)

Standard Metadata: Committee: ETD 20 (Electrical Installations) | Scope: Design, sizing, installation, and testing of protective earthing, system earthing, earth electrodes, substation grounding grids, and touch/step potential mitigation.

5.1 Fundamental Objectives & Safety Limits

Earthing ensures human safety by limiting potential differences across accessible conductive parts, preventing electric shock, providing low-impedance return paths for fault current to trigger overcurrent protective devices (MCBs/MCCBs/Relays), and dissipating lightning discharges.

- **Permissible Earth Resistance Thresholds:**
 - Large Generating Stations & EHV Substations (400 kV / 220 kV): $\leq 0.5 \Omega$
 - Medium High-Voltage Substations (33 kV / 11 kV): $\leq 1.0 \Omega$
 - Industrial Plants, Data Centers, and Main Distribution Boards: $\leq 1.0 \Omega$
 - Commercial & Domestic Low-Voltage Consumer Premises: $\leq 5.0 \Omega$ (Ideally $< 2.0 \Omega$ for sensitive electronics).
- **Step and Touch Voltage Hazards:**
 - *Touch Voltage* (E_{touch}): Potential difference between a grounded metal enclosure and the human feet standing 1 m away during a line-to-ground fault.
 - *Step Voltage* (E_{step}): Potential difference between two feet separated by 1 m on the soil surface without touching any structure.

5.2 Soil Resistivity (ρ) & Earthing Calculations

Soil resistivity ρ (measured in $\Omega \cdot m$) is measured using the Wenner 4-pin method: $\rho = 2 \pi s R$, where s is pin spacing and R is measured resistance. Typical ranges: Garden loam ($10 - 50 \Omega \cdot m$), Clay ($20 - 100 \Omega \cdot m$), Sandy loam ($50 - 250 \Omega \cdot m$), Rocky/Dry granite ($1000 - 10000 \Omega \cdot m$).

• **Vertical Pipe / Rod Electrode Resistance (R_R — Clause 14.2.2):**

$$R = \left[\frac{100 \cdot \rho}{2 \cdot \pi \cdot L} \right] \cdot \ln \left(\frac{2 \cdot L}{d} \right) [\Omega]$$

where ρ is soil resistivity in $\Omega \cdot m$, L is rod buried length in cm, and d is electrode outer diameter in cm.

• **Horizontal Strip / Conductor Electrode Resistance (R_R — Clause 14.2.3):**

$$R = \left[\frac{100 \cdot \rho}{2 \cdot \pi \cdot L} \right] \cdot \ln \left(\frac{4 \cdot L}{d} \right) [\Omega]$$

where L is strip length in cm and d is width of strip or twice wire diameter in cm.

5.3 Physical Dimensions & Specifications of Electrodes

Electrode Type	Material	Minimum Dimensions / Cross-Section	Mandatory Installation Details
Plate Electrode	Galvanized Iron (GI) / Steel	600 mm imes 600 mm imes 6.3 mm	Buried vertically with top edge ≥ 1.5 m below ground level.
Plate Electrode	Copper	600 mm imes 600 mm imes 3.15 mm	Buried vertically with top edge ≥ 1.5 m below ground level.
Pipe Electrode	Galvanized Steel / Cast Iron	Internal Diameter ≥ 38 mm (GI) or ≥ 100 mm (CI); Length ≥ 2.5 to 3.0 m	Perforated holes along length; surrounded by bentonite or charcoal-salt layers.
Rod Electrode	Copper-bonded steel / Solid Copper	Diameter ≥ 16 mm (steel), ≥ 12.5 mm (copper); Length ≥ 2.5 m	Driven vertically into undisturbed ground.
Strip / Wire Electrode	GI or Copper Strip	25 mm imes 4 mm (GI) or 25 mm imes 1.6 mm (Copper)	Buried in horizontal trenches at depth ≥ 0.5 m (preferably $0.8 - 1.0$ m).

5.4 Earthing Conductor (Fault Current Sizing — Clause 12.2)

The minimum cross-sectional area S of earthing conductors carrying fault current without exceeding permissible thermal limits:

$$S = (I \cdot \sqrt{t}) / K [mm^2]$$

where I is RMS fault current in Amperes, t is fault clearing duration in seconds (typically 1.0 s or 0.5 s), and K is material temperature constant ($K = 80$ for galvanized steel, 159 for copper with PVC insulation, 226 for bare copper).

RAG Keyword Vector Seed: IS 3043:2018, earthing code, earth resistance 1 ohm substation 5 ohm domestic, soil resistivity ρ , vertical rod formula $\ln(2L/d)$, plate electrode $600 \times 600 \times 6.3 \text{ mm}$ GI, copper plate 3.15 mm , burial depth 1.5 m , $S = (I \sqrt{t})/K$.

6. IS 10500 : 2012 — Drinking Water - Specification (Second Revision)

Standard Metadata: Committee: FAD 14 (Drinking Water) | Scope: Quality parameters, sensory traits, chemical toxins, radiological criteria, and bacteriological requirements for potable water.

6.1 Mandatory Two-Tier Threshold Philosophy

IS 10500:2012 classifies water quality limits into:

- **Acceptable Limit:** Target value for water to be completely safe, wholesome, and aesthetically pleasing for domestic consumption.
- **Permissible Limit in the Absence of an Alternate Source:** Maximum allowable ceiling in drought-affected or geographically constrained locations where alternate water supplies are unavailable. *Exceeding this permissible limit warrants mandatory source condemnation.*

6.2 Comprehensive Quality Table (Physical, Chemical & Toxic Contaminants)

Test Parameter	Unit	Acceptable Limit	Permissible Limit (No Alternate)	Testing Standard Reference	Health & Aesthetic Impact
Color	Hazen Units	5	15	IS 3025 (Part 4)	Aesthetic unacceptability
Turbidity	NTU	1.0	5.0	IS 3025 (Part 10)	Inhibits chlorine disinfection; pathogen shelter
pH Value	—	6.5 – 8.5	No relaxation	IS 3025 (Part 11)	\$< 6.5\$ corrosive to pipes; \$> 8.5\$ mineral scaling
Total Dissolved Solids (TDS)	mg/L	500	2000	IS 3025 (Part 16)	Bitter taste; laxative for unaccustomed consumers
Total Hardness (as CaCO3)	mg/L	200	600	IS 3025 (Part 24)	Scale in boilers/geysers; high soap consumption
Alkalinity (as CaCO3)	mg/L	200	600	IS 3025 (Part 23)	Unpleasant taste; reaction with water fittings
Chloride (as Cl)	mg/L	250	1000	IS 3025 (Part 32)	Salty taste; high metal corrosion rate
Sulphate (as SO4)	mg/L	200	400	IS 3025 (Part 24)	Gastrointestinal irritation; laxative effect
Fluoride (as F)	mg/L	1.0	1.5	IS 3025 (Part 60)	\$< 1.0\$ dental caries; \$> 1.5\$ dental & crippling skeletal fluorosis
Nitrate (as NO3)	mg/L	45	No relaxation	IS 3025 (Part 34)	Causes infantile Methemoglobinemia (Blue Baby Syndrome)
Iron (as Fe)	mg/L	0.3	No relaxation (or 1.0)	IS 3025 (Part 53)	Stains plumbing fixtures/laundry; metallic taste
Arsenic (as As)	mg/L	0.01	0.05	IS 3025 (Part 37)	Class 1 Carcinogen; skin keratosis, internal cancers
Lead (as Pb)	mg/L	0.01	No relaxation	IS 3025 (Part 47)	Neurotoxin; cognitive impairment, kidney disease
Total Chromium (as Cr)	mg/L	0.05	No relaxation	IS 3025 (Part 52)	Ulcerations; organ toxicity
Total Coliform Bacteria	MPN / 100 mL	Must not be detectable	No relaxation	IS 1622	Fecal contamination; cholera, dysentery, typhoid
Escherichia coli (E. coli)	/ 100 mL	Must not be detectable	No relaxation	IS 1622	Immediate indication of active mammalian fecal pollution

RAG Keyword Vector Seed: IS 10500:2012, drinking water, pH 6.5-8.5, TDS 500 acceptable 2000 permissible, Turbidity 1 to 5 NTU, Fluoride 1.0 to 1.5 mg/L, Nitrate 45 mg/L no relaxation, Arsenic 0.01 mg/L, Lead 0.01 mg/L, E coli zero coliform absent.

7. IS 13252 (Part 1) : 2010 / IS/IEC 60950-1 — Information Technology Equipment - Safety

Standard Metadata: Committee: LITD 7 (IT Hardware & Audio/Video) | Scope: Product electrical, thermal, mechanical, radiation, and fire safety requirements for IT equipment under MeitY Compulsory Registration Scheme (CRS).

7.1 Scope, Energy Hazards & Circuit Classifications

IS 13252 (Part 1) establishes statutory safety mandates for mains-connected or battery-operated information technology hardware (laptops, power adapters, servers, network switches, POS terminals, and printers). It systematically isolates users and service personnel from electric shock, excessive temperatures, fire hazards, and mechanical instability.

• Circuit Voltage Categories (Clause 1.2.8 & 1.4):

- **SELV Circuit (Safety Extra-Low Voltage):** Secondary circuit with protective separation where voltage between any two conductors (or conductor to ground) does not exceed $42.4 \text{ ext{ V}} \text{ peak AC or } 60 \text{ ext{ V DC}}$ under normal operating conditions or single fault conditions. Accessible to users.
- **ELV Circuit (Extra-Low Voltage):** Circuit voltage within SELV limits, but lacking full isolation guarantees from hazardous voltages.
- **Hazardous Voltage:** Any voltage exceeding $42.4 \text{ ext{ V}} \text{ peak AC or } 60 \text{ ext{ V DC}}$ in circuits without current-limiting protection.

7.2 Insulation, Clearance & Creepage Distances (Clause 2.9 & 2.10)

- **Clearance:** The shortest distance through air between two conductive parts. Prevents dielectric breakdown (sparkover) under transient overvoltages.
 - For primary mains circuit ($250 \text{ ext{ V RMS}}$), $2.5 \text{ ext{ kV}}$ impulse): Minimum clearance for basic insulation = $1.5 \text{ ext{ to } } 2.0 \text{ ext{ mm}}$; for reinforced insulation = $3.0 \text{ ext{ to } } 4.0 \text{ ext{ mm}}$.
- **Creepage Distance:** The shortest path along the physical surface of solid insulating material between two conductive parts. Prevents tracking failure due to moisture and dust deposition.
 - For $250 \text{ ext{ V RMS}}$ in Pollution Degree 2, Material Group III: Basic insulation requires $\geq 2.5 \text{ ext{ to } } 3.0 \text{ ext{ mm}}$; Reinforced insulation requires $\geq 5.0 \text{ ext{ to } } 6.0 \text{ ext{ mm}}$.

7.3 Protective Earthing, Touch Leakage Current & Dielectric Tests (Clause 5)

- **Touch Leakage Current Thresholds (Clause 5.1 & Table 5A):**
 - Handheld portable IT equipment (laptops, mobile chargers): Max $0.75 \text{ ext{ mA}}$.
 - Movable non-handheld IT equipment: Max $3.5 \text{ ext{ mA}}$.
 - Stationary pluggable Type A equipment: Max $3.5 \text{ ext{ mA}}$.
 - Stationary permanently wired or industrial rack servers: Max 5.0% of input current (with warning label and high-integrity earth).
- **High-Voltage Dielectric Withstand Test (Hi-Pot Test — Clause 5.2):** Equipment rated for $220 \text{ ext{--} } 250 \text{ ext{ V}}$ AC mains must withstand an insulation test voltage without flashover or insulation breakdown for **60 seconds**:
 - Basic Insulation: $1500 \text{ ext{ V RMS}}$ AC (or $2121 \text{ ext{ V DC}}$).
 - Reinforced / Double Insulation: $3000 \text{ ext{ V RMS}}$ AC (or $4242 \text{ ext{ V DC}}$).

RAG Keyword Vector Seed: IS 13252, IEC 60950-1, SELV 42.4V peak 60V DC, clearance 2.0mm, creepage 2.5mm basic 5.0mm reinforced, touch current 0.75mA handheld 3.5mA movable, dielectric test 1500V AC 3000V reinforced 60s, MeitY CRS registration.

8. IS 383 : 2016 — Coarse and Fine Aggregate for Concrete - Specification (Third Revision)

Standard Metadata: Committee: CED 2 (Cement & Concrete) | Scope: Quality, grading, soundness, mechanical strength, and deleterious contamination thresholds for coarse and fine aggregates including natural, crushed, and manufactured sands (M-Sand).

8.1 Scope & Inclusion of Alternative Aggregate Sources

IS 383:2016 introduced provisions permitting manufactured fine aggregates (**M-Sand**), blast furnace slag aggregates, copper slag, and Recycled Concrete Aggregates (RCA), directly addressing environmental restrictions on natural river sand extraction.

8.2 Fine Aggregate Grading Zones (Clause 6.3 & Table 9)

Sand is classified into four discrete grading zones based on sieve analysis and particle size distribution. The **600 μm sieve** is the primary diagnostic sieve separating the zones:

IS Sieve Size	Grading Zone I	Grading Zone II	Grading Zone III	Grading Zone IV
10 mm	100%	100%	100%	100%
4.75 mm	90 – 100%	90 – 100%	90 – 100%	95 – 100%
2.36 mm	60 – 95%	75 – 100%	85 – 100%	95 – 100%
1.18 mm	30 – 70%	55 – 90%	75 – 100%	90 – 100%
600 μm (Key Sieve)	15 – 34%	35 – 59%	60 – 79%	80 – 100%
300 μm	5 – 20%	8 – 30%	12 – 40%	15 – 50%
150 μm	0 – 10%	0 – 10%	0 – 10%	0 – 15%

- Zone I (Coarse Sand):** Fineness modulus 3.5 to 4.0. Used for lean mass concrete and pavement sub-bases.
- Zone II (Standard Concrete Sand):** Fineness modulus 2.9 to 3.5. **The primary specification for structural reinforced concrete work (RCC).**
- Zone III (Medium-Fine Sand):** Fineness modulus 2.3 to 2.9. High pumpability concrete and external plastering.
- Zone IV (Very Fine Sand):** Fineness modulus 1.5 to 2.1. Prohibited for reinforced structural concrete; used for internal wall plastering and masonry mortar.

8.3 Physical, Mechanical & Deleterious Substance Limits (Clause 5)

- Deleterious Material Limits (Table 1):**
 - Clay Lumps: Maximum **1.0%** by mass in both fine and coarse aggregate.
 - Fines passing 75 μm sieve:
 - Natural uncrushed river sand: Maximum **3.0%** by mass.
 - Crushed stone sand / Manufactured Sand (M-Sand): Permitted up to **15.0%** by mass (provided fineness is micro-fine stone dust rather than expansive clays).
 - Coal and Lignite: Maximum **1.0%** by mass.
- Flakiness and Elongation Index (Clause 5.2):** Combined flakiness and elongation index of coarse aggregates must not exceed **40%** (or **35%** for wearing surfaces/airfields).
- Mechanical Strength Thresholds (Clause 5.4):**
 - Aggregate Crushing Value (ACV):* Maximum **30%** for concrete wearing surfaces; maximum **45%** for general structural concrete.
 - Aggregate Impact Value (AIV):* Maximum **30%** for concrete wearing surfaces; maximum **45%** for general structural concrete.
 - Los Angeles Abrasion Value:* Maximum **30%** for wearing surfaces; maximum **50%** for general structural concrete.
 - Soundness Test (5 cycles):* Loss in mass with Sodium Sulphate (Na_2SO_4) $\leq 10\%$ for fine aggregate, $\leq 12\%$ for coarse aggregate.

RAG Keyword Vector Seed: IS 383:2016, aggregates specification, M-Sand, 600 micron sieve, Zone I 15-34%, Zone II 35-59% RCC, Zone III 60-79%, Zone IV 80-100% plaster, clay lumps 1%, silt 3% natural 15% M-sand, crushing value 30% wearing 45% non-wearing.